

Activity Report

"Orientation on Hollow Concrete Block Manual"

Ramechhap District

12th March, 2019

Activity Date and Venue:	03/12/2019, GS Hotel, Manthali, Ramechhap
Guests	Mr. Krishnalal Piya(GMAI Chief Ramechhap)
Facilitator	Er. Tulasi Ram Lamichhane (HRRP Ramechhap), Jawed Mohammad (HRRP, Ramechhap)
Report Prepared By:	Tulasi Ram Lamichhane

Discussion:

Welcome Remarks and Introduction

- The orientation program was facilitated by District General Coordinator of HRRP Ramechhap, Prakash Bishwakarma. He welcomed all the participants in the Orientation on Hollow Concrete Manual. He briefed that the orientation program was targeted to the technical staffs of DLPIU, Rural/Municipality and Partner Organizations who have been working in housing sector in Ramechhap District. The program began with the brief introduction of all participants and resource person.

Hollow Concrete Block Manual (Er. Tulasi Ram Lamichhane, HRRP Ramechhap)

The hollow Concrete Block (HCB) Manual for Load Bearing Structures, which has been published by National Reconstruction Authority (NRA) has been targeted to support the construction of the houses using hollow concrete blocks in the earthquake affected districts. The manual is a written guideline for the engineers to provide their technical assistance for the house owners who prefer hollow concrete blocks for the reconstruction.

The Hollow Concrete Block Manual has been divided into six parts and five annexes:

PART 1: Background

PART 2: Building Typology and Inspection

PART 3: Technical Inspection

PART 4: Quality Control Vs. Assurance

PART 5: Correction Measures

PART 6: Ready to Use Designs

Annex A: HCB Unit Quality Test at Site

Annex B: Structural Analysis and Design

Annex C: Estimate of Correction Measures

Annex D: Case Study on Inspection

Annex E: Inspection Forms (HCB Masonry)

The HCB Manual has been prepared based on NBC105:1994, NBC202:1994 and IS 2185:2005 and is applicable for the load bearing structures under the GoN reconstruction program. It has been introduced to provide the grant to the houses constructed using the Hollow Concrete as the structural element.

The concrete blocks are classified as Hollow concrete block and solid concrete block based on the volume occupied by the concrete in the block. The size of the concrete blocks has been defined as 400mmx200mmx(100mm/150mm/200mm) for the full block and 200mmx200mmx(100mm/150mm/200mm) for the half block. The concrete blocks have several advantages over other construction materials if the quality of the block is assured.

The construction sequence of a load bearing structure using hollow concrete block has been elaborated in an algorithm with the specifications of each elements which can be considered as the minimum requirements. The ready to use design included in the manual is applicable for the structure with following specifications:

- No. of storey: 2
- Total height of storey :2.5 m (maximum)
- Total building Height: 5 m
- Unsupported wall length: 4.05 m (maximum)
- Plinth area : 100.00 sq.m.

IM Session (Jawed Mohammad, HRRP Ramechhap)

IM session was focused on Action Plan and Palika Profile of Ramechhap District. Briefly discussed on the importance of the Action Plan in the following topics.

- १.१ निजी आवास पुनर्निर्माणको वर्तमान अवस्था र लक्ष्य
- १.२. गुनासो व्यवस्थापन
- १.३. जौखिम बस्ती स्थानान्तरण
- १.४. एकीकृत वस्ती
- १.५. सामुदायिक समिति गठन
- १.६. कामदारको आपूर्ती, निर्माण सामाग्रीको आवश्यकता, आपूर्ती तथा बजारको अवस्था
- १.७. प्राविधिक जनशक्तिको उपलब्धता
- १.८ समस्या / चुनौति र समाधानका उपायहरु

Also brief to the focal person on Palika Profile which gives overall reconstruction glimpse: damage status, tranche update for both full reconstruction and retrofitting, grievance status, non compliance issues, type of housing typology, partner organizations in Palika including active and non active, construction material required for reconstruction of houses along with current rate, land issues, technical staff status, training (short and vocational) status, seven TA Maps, frequently asked questions, contact details of reconstruction focal offices, which is resourceful document to the Palika.

At last also request focal person to update the Action Plan in the monthly basis.

Closing Remarks:

- **Mr. Krishnalal Piya (Chief GMaLI Ramechhap)**

Mr. Krishnalal Piya appreciated HRRP in conducting the program and briefed that the new construction materials in reconstruction will rapid the reconstruction as the construction time needed is less. Though the hollow concrete structures have not been reported in the district, this program will build the capacity of the technical persons of reconstruction. This program will help the technical persons to implement such structures and study about them in future. If anyone needs technical assistance regarding such structures, the technical person will be able to assist the house owners.

Questions Collected	Who asked the Question?	Response Provided
What is the minimum width of the wall for 2 storey house of hollow concrete block	DLPIU	The manual doesn't specify the exact width for the house based on storey. It states the minimum width must be 150mm. But In case of 2 storey building we can increase the width of the building via its structural analysis.
Why is the strength of unit greater than that of the wall	GMALI	This is because of the stiffness, joints and other factors which reduces the strength of wall because of its height and different material composition.

Participants Detail						
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Guests and Resource person

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<i>District Coordinator of Ramechhap facilitating the orientation program on Hollow Concrete Block Manual.</i>	<i>District Technical Coordinator of Ramechhap presenting the slides of orientation on Hollow Concrete Block Manual.</i>
	
<i>IMO of HRRP Ramechhap presenting on Palika Level Action Plan.</i>	<i>NRA/GMaLI Chief giving his closing remarks on Orientation on Hollow Concrete Block Manual.</i>